



Comparing Canopy Height Models from Regional-Scale Aerial Photogrammetry with Global Spaceborne Lidar-Derived Data for Estimating Forest Volume and Biomass

Qianqian Cao¹ · Philip J. Radtke¹ · John W. Coulston² · Valerie A. Thomas¹ · Randolph H. Wynne¹ · David M. Walker³

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Abstract

Forest inventory practitioners require information at finer geographic scales than ever to provide information for assessing forest sustainability, carbon storage, and other policy and management needs. Reliable assessments of estimation uncertainties and timely delivery of information are also increasingly critical. Small area estimation (SAE) techniques have been successfully applied in forest inventory to address these needs, borrowing strength from broad scale inventory data paired with auxiliary information – often remote sensing – to improve the precision of sample-based estimates. This work presents a comparative analysis of SAE results using auxiliary canopy height information from two remote-sensing sources, both canopy height models (CHM), one derived from digital aerial photogrammetry and existing digital elevation models, the other from spaceborne lidar and multitemporal surface-reflectance observations. Area-level Fay-Herriot models were developed using field observations from the US National Forest Inventory (NFI) applied to three states in the southeastern US. Supplemental filtering of CHMs to align with the NFI's definition of forest was explored. Precision was increased in county-level estimates of both forest wood volume and live tree aboveground biomass using either of the CHM products, with greater precision gains possible by excluding remote-sensing data from non-forested areas. Results demonstrated precision gains possible from incorporating state-wide CHM data from either aerial or spaceborne remote sensing platforms with NFI sample data to estimate timber volume and biomass for counties in three eastern states in the USA. Further work is needed to determine where synthetic model predictions without random effects might serve in place of composite Fay-Herriot estimators.

Keywords Global ecosystem dynamics investigation (GEDI) · National agricultural imagery program (NAIP) · Fay-herriot · Small area estimation

Extended author information available on the last page of the article

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Introduction

National forest inventories (NFI) are highly effective for accurately assessing forest resources at broad geographic scales including nations, states, or provinces, largely due to their rigorous survey designs and extensive networks of sample field data. As an example, the United States NFI, directed by the USDA Forest Service, Forest Inventory and Analysis (FIA) program, specifies an allowable error of $\pm 5\%$ (one standard error) per estimated 28.3 million m³ (1 billion cubic feet) of wood volume on commercial forest land in 31 eastern states, i.e. those bordering the Mississippi River or entirely east of it (USDA Forest Service National Headquarters 2008).

In contrast, NFIs may be limited in their ability to produce highly accurate estimates at finer spatial scales such as in counties or municipalities, due to the relative paucity of sample data available in smaller areas. For instance, only 101 of 1,995 counties in the FIA inventory of eastern states met the 28.3 million m³ volume threshold (USDA Forest Service Forest Inventory and Analysis Program 2022). Consequently a 5% allowable error for wood volume estimates should not be considered a realistic goal for most eastern US counties.

To avoid instability and large variance associated with estimating attributes from sparse sample data, small area estimation (SAE) has been studied increasingly in forest inventory research over the past few decades (Dettmann et al. 2022). Area-based SAE using the widely-adopted Fay-Herriot (FH) estimator (Fay and Herriot 1979) has been demonstrated in forest inventory research for counties, municipalities, and other small geographic areas (Breidenbach et al. 2018; Temesgen et al. 2021). A notable concern with the FH method is the estimation of random area effects for areas of interest (AOI) when the area-effect variance is small or potentially zero (Molina et al. 2015). A number of estimators for area-effect variance have been proposed to avoid undesirable results such as negative values or biased estimates (Li and Lahiri 2010; Yoshimori and Lahiri 2014). Available FH variance estimators may give values that are negative, zero, or strictly positive, with negative estimates typically set to zero in statistical software. The choice of variance estimator can affect both the apparent gains realized from adopting SAE methods as well as the ability to estimate random effects used to formulate FH composite estimates.

Linking auxiliary information to sample data when observations from the two are correlated has been key to accelerating the development of SAE methods and applications (Pfeffermann 2002, 2013; Sugasawa and Kubokawa 2020). In forest inventory, it has been widely recognized that remote sensing data sets – often obtained from aerial or spaceborne platforms – provide auxiliary information that can supplement observations from field plots (Breidenbach and Astrup 2012; Goerndt et al. 2013; McRoberts et al. 2013). Significant gains in precision are possible from such efforts while preserving favorable properties of consistency and unbiasedness in forest inventory estimators (Dettmann et al. 2022).

Past work has demonstrated the utility of forest canopy height models (CHM) from aerial remote sensing as auxiliary information to increase precision of forest

inventory estimates in small geographic AOI (Cao et al. 2022; Magnussen et al. 2016). While aerial laser scanning (ALS), i.e. lidar, has drawn considerable attention for such uses (Babcock et al. 2018; Breidenbach and Astrup 2012; Goerndt et al. 2011; Mauro et al. 2019), point clouds derived from digital aerial photogrammetry (DAP) have also proven effective as sources for developing CHMs for use in forest inventory applications (Prior et al. 2022; Schroeder et al. 2022; Strunk et al. 2020).

Recent results from the NASA Global Ecosystem Dynamics Investigation (GEDI) satellite mission led to the development of a 30 m global forest canopy height model (GFCHM) derived from spaceborne lidar (Dubayah et al. 2020; Potapov et al. 2021). While the GEDI-derived GFCHM was not released as an official GEDI science data product, its reliance on the sensor's elevation, height, canopy cover, and vertical profile metrics, together with a large collection of ALS data and Landsat optical time-series data for land cover mapping and change detection, make it a strong candidate for use as auxiliary data in forest inventory SAE applications (Potapov et al. 2020).

Here we build on the work of Cao et al. (2022) that showed strong gains in sampling efficiency when pairing DAP-derived CHM data with NFI sample data to estimate total wood volume and aboveground biomass (AGB) for counties in three eastern states in the USA. A primary goal is to compare precision gains for SAE conducted using satellite-based GFCHM as auxiliary data to gains using DAP-derived CHMs from aerial imagery collected under the USDA National Agriculture Imagery Program (NAIP). Since neither the GEDI-derived GFCHM nor NAIP-derived CHM products were designed to target areas of forest land use exactly as defined by FIA, a secondary goal here is to examine whether any *a priori* filtering of CHMs to better align with forest land delineation under different definitions can improve gains in SAE precision. We also examine any differences in the performance of SAE comparing the two CHM data sources for estimating forest wood volume and AGB in live trees. Our final objective is to examine how an estimator of random effects variance adjusted to ensure non-negativity affects SAE county level inventory estimates, particularly with respect to hypothesis testing for zero random effects variance. We anticipate that our results will inform practitioners' decisions for using either GEDI-derived GFCHM or NAIP 3d DAP CHMs in forest inventory area-level SAE, and consideration for choosing from alternative variance estimation approaches when facing uncertainty over the need to account for area effects.

Methods

Study Area and Sample Data

The study area included three states of North Carolina (NC), Tennessee (TN), and Virginia (VA) in the southeastern USA (Table 1). States were chosen based on the availability of remote sensing CHM data from both DAP-derived NAIP imagery and the GEDI satellite-derived GFCHM (Cao et al. 2022; Potapov et al. 2021). Political subdivisions within the states divide them into 95 counties in TN, 99 counties in VA (excluding Arlington, which had no forest plots within its boundaries), and 100 in

Table 1 Forest volume, above ground biomass (AGB), and land area statistics for states in the study area, with the number of forested FIA sample plots (n) in each state

State	Net volume ($\text{m}^3 \times 10^6$)	AGB ($\text{Mg} \times 10^6$)	Land area (km^2)	Forest area (%)	No. forest plots	Minimum No. Forest plots/county	Maximum No. Forest plots/ county
NC	1,251	990.7	139,400	54	3,662	6	80
TN	926	758.1	109,200	52	2,932	2	94
VA	1,145	915.7	110,800	62	3,298	1	86

NC, for an average county land area of 1220 km² for the 294 counties in the study area (Table 1).

Direct estimates of both wood volume and AGB in forests were obtained from the FIA program database of sample observations for the United States' NFI (USDA Forest Service FIA 2021). Estimates were based on the FIA calendar year 2017 for forest evaluation for all states (the most recent year estimates were available for all three states at the time the work was pursued), so that full-panel estimates collected over 5–7 years could be used in estimation (Bechtold and Patterson 2005). By design, full-panel estimates from FIA exhibit lower sample variance than estimates from partial-panel data such as those collected in a single calendar year. Due to an assumption of temporal indifference in FIA full-panel estimators, slight biases may exist in estimates for individual years within the multi-year period; however, we favored the greater sampling intensity from full-panel information in the trade-off between bias and precision of direct inventory estimates.

Sample plot observations were used to construct estimates of forest volume and AGB, along with standard errors, for each county within the three states (Table 1). County total forest wood volume was defined for live tree main stems having diameters at breast height (dbh) ≥ 12.7 cm (i.e., 5.0 inch), excluding wood in stumps below a 30.5 cm (1 foot) height and stem topwood above a 10.2 cm (4 inch) upper stem diameter. AGB in each county was defined as the oven-dry biomass in the above-ground portion of live trees with dbh ≥ 2.54 cm (i.e., 1.0 inch). Roots, foliage and wood lost to mortality, damage or decay were not included in AGB. For trees having dbh at least 12.7 cm (i.e., 5 inch), AGB estimated by FIA uses observed species, DBH, and tree heights following the component ratio method (CRM; Heath et al. 2009). AGB for smaller trees known as saplings ($2.54 \text{ cm} \leq \text{dbh} < 12.7 \text{ cm}$) is computed using equations of Jenkins et al. (2003). Details of volume and biomass determination for individual trees in the FIA database are described by Woodall et al. (2011). The estimated wood volumes and AGB for each county and their respective sample variances were obtained following documented estimation procedures for the FIA database (Bechtold and Patterson 2005; Pugh et al. 2018).

Remote Sensing Data

NAIP-derived CHMs were generated from aerial photography acquired in coordination with the NAIP program under state and federal partnerships aimed at evaluating the suitability of 3 d-DAP products for improving statewide forest inventory estimates. GFCHMs were downloaded from the University of Maryland Global Land Analysis and Discover (GLAD) repository at glad.umd.edu/dataset/gedi. Acquisition years of CHM data from NAIP imagery and GFCHM were 2018 and 2019, respectively. Data from NAIP were delivered as digital surface model (DSM) raster files having grid cells of approximately 1×1 m (TN) and 5×5 m (NC and VA) produced from 3 d-DAP (Lu et al. 2016; Strunk et al. 2020). The DSM data were resampled at 10×10 m resolution for overlay with US National Elevation Database Digital Elevation Model (DEM) data (US Geological Survey 2019). CHMs were

calculated by subtracting DEM values from the NAIP 10 m DSMs, setting any negative values to zero.

Potapov et al. (2021) derived the GFCHM data from multiple remote sensing sources including NASA's Landsat 30-m resolution multi-spectral time-series imagery and the GEDI footprint-based waveform lidar instrument operating onboard the International Space Station since April 2019. The Landsat time series data provided a basis for Potapov et al. (2021) to impute forest tree canopy heights from the relatively sparse GEDI footprint samples to nearly complete spatial coverage across much of the Earth.

CHM raster layers from both NAIP and GFCHM sources were clipped to each county boundary and aggregated into seven 5-m interval height classes spanning (0, 5], (5, 10], ..., (30–35] m, with values outside the range (0, 35] m omitted to remove possible outliers, anomalies, or other errors, assuming nearly all forest canopies in the study area were ≤ 35 m in height. County areas covered by each of the seven height classes or “bins” denoted (B5, B10, ..., B35) were computed as the product of grid cell counts and cell area, converted to km², with each bin variable evaluated for efficacy in linear regression models for predicting county totals of both forest wood volume and AGB.

Auxiliary Information Filtering

In comparing NAIP and GFCHM data for use as auxiliary information in SAE, we set out to rectify differences in how any filtering was applied during development to limit CHM coverage to forested or tree-canopied areas. Forest wood volume and AGB are both highly dependent on the land area occupied by forests in counties of interest. As such, we expected the strength of relationships between CHM predictors and FIA estimates would be affected by any filtering applied to CHM data sets related to forest area classification. As delivered, NAIP DSMs made no accounting for forests or other land cover types, including water or urban areas. In contrast, the available GFCHM data were designed to quantify only those height measurements that could be directly attributed to tree or forest canopies. To accomplish this goal, Potapov et al. (2021) applied a series of filters to remove pixels classified as buildings, urban areas, agricultural crops, or other classification types that indicated potential for large errors. Neither NAIP nor GFCHM data were originally intended to apply to areas of forest land use exactly as defined by FIA for its estimates; therefore, we applied our own filters to both remote-sensing data sets to better align FIA's forest land-use definition with forest cover classifications based on remote sensing (Coulston et al. 2014). This alignment was designed to ensure comparability among the alternative datasets.

The first filter we applied to NAIP CHMs was the removal of areas classified as water in the 2019 National Land Cover Database (Dewitz and USGS 2021). This filter – denoted as “NAIP_noWater” (Fig. 1) – was needed to remove notable errors related to DAP height measurements over open water (Cao et al. 2022). The second filter we applied to NAIP CHMs was the transfer of all zero values from the GFCHM raster dataset to corresponding pixels in the NAIP coverage, which we denoted

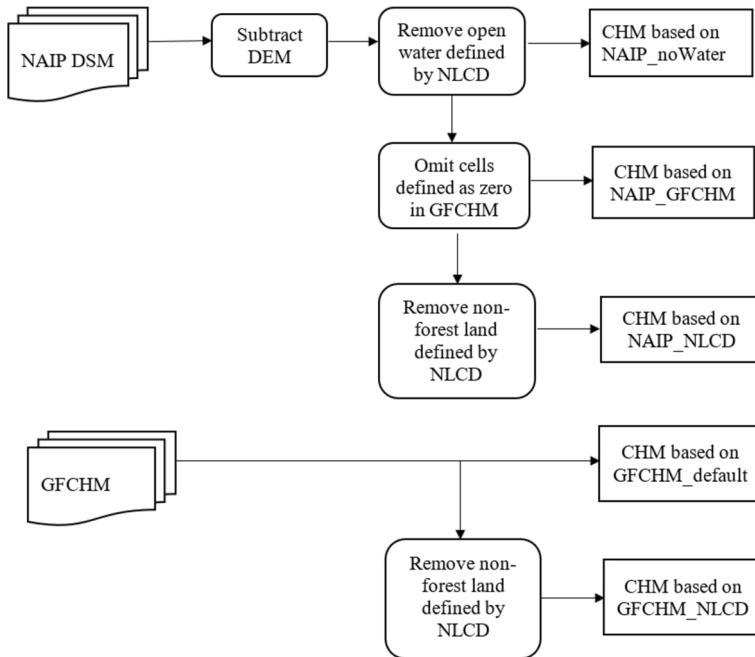


Fig. 1 Filters tested for remote-sensing data based from NAIP and GFCHM canopy height data for identifying filters that give optimal results for estimating forest wood volume and aboveground biomass

as “NAIP_GFCHM” (Fig. 1). Our expectation was that the NAIP_GFCHM filter would improve comparability between NAIP and unmodified (i.e. default) GFCHM data as downloaded, even though neither would strictly follow the FIA definition of forest land use. Third, both data sets were filtered to remove CHM values that did not coincide with NLCD 2019 land cover classes for deciduous, evergreen, or mixed forests, woody wetlands, or shrub/scrub, which we denoted as NAIP_NLCD and GFCHM_NLCD for NAIP and GFCHM datasets, respectively. The intent in this step was to maintain comparability while more closely matching the FIA definition of forests – including regenerating forests typically included in NLCD shrub/scrub and woody wetland classes in eastern states. We assumed this filter might better coincide with NLCD forest cover classes than the slightly broader tree canopy classification adopted by the GFCHM product.

SAE Area-Level Fay-Herriot Model

Area-level SAE models, including widely-used FH estimators (Fay and Herriot 1979), use a weighted-average composite of direct and indirect estimates that result in empirically best linear unbiased predictions (EBLUP) of attributes of interest in each small area domain. In this study, counties were treated as small area domains. Forest wood volume and AGB were attributes of interest, with the population total for either attribute

in a given county (d) denoted as τ_d . Analyses were conducted using the R package ‘sae’ (Molina and Marhuenda 2015b), which implements the composite FH estimator

$$\hat{\tau}_d^{EBLUP} = \hat{\gamma}_d \hat{\tau}_d^{DIR} + (1 - \hat{\gamma}_d) X_d^T \hat{\beta} \quad (1)$$

where $\hat{\tau}_d^{DIR}$ is the direct estimate of total volume (million m³) or AGB (million metric tons, also Mg) for the d^{th} county computed using sample observations and design weights from FIA. The vector X_d consists of the canopy height class bin areas (e.g., B5, B10, ..., B35) in km² for county d as determined from a CHM, with $\hat{\beta}$ the set of linear fixed-effect regression coefficients estimated from $d = 1, 2, \dots, D$ county-level observations of $X = X_d$ and $Y = \hat{\tau}_d^{DIR}$ in a particular state. The composite weighting factor, i.e. shrinkage $\hat{\gamma}_d$ was defined as

$$\hat{\gamma}_d = \frac{A}{(A + \hat{v}(\hat{\tau}_d^{DIR}))} \quad (2)$$

where v denotes the variance of the direct estimate and $A = \hat{\sigma}_a^2$ is the estimated variance among the area (county) random effects specified in a reformulation of [1] as a linear mixed-effects regression model (Goerndt et al. 2011). Shrinkage [2] provides a way to balance desirable properties of design-unbiasedness in direct estimates $\hat{\tau}_d^{DIR}$ and precision of regression-synthetic estimates $X_d^T \hat{\beta}$. In model [1], the shrinkage factor [2] accounts for relative sizes of the sample-direct estimator variance $\hat{v}(\hat{\tau}_d^{DIR})$ within counties as well as the regression variance among counties measured by the random effect variance A . Regression coefficients and component variances were estimated using restricted maximum likelihood (REML) in the ‘sae’ package. Additional details of the estimation procedure are given by (Molina and Marhuenda 2015b).

Adjusted Likelihood Estimator of Random Effect Variance

Variance estimation using REML can produce negative estimates of random effect variance A , especially when A is small relative to sample-direct estimate variances (Molina and Marhuenda 2015a). Negative estimates for A are set to zero in most FH fitting algorithms (Li and Lahiri 2010; Yoshimori and Lahiri 2014). Consequently, the shrinkage term $\hat{\gamma}_d$ in [1] is set to 0 when $A = 0$. This behavior, sometimes referred to as over-shrinkage, reduces EBLUPs in [1] to regression synthetic model predictions $X_d^T \hat{\beta}$ for all domains. Over-shrinkage may interfere with the FH model EBLUPs’ ability to balance properties of unbiasedness from sample-direct estimates with precision of regression synthetic estimates. To avoid zero estimates of A , adjustment factors based on maximum likelihood have been developed (Li and Lahiri 2010; Wang and Fuller 2003; Yoshimori and Lahiri 2014). We used the adjusted residual maximum likelihood calculation $L_{adj}(A)$ proposed by Yoshimori and Lahiri (2014) to obtain strictly non-negative random effect variances.

$$L_{adj}(A) \propto h(A) \times L(A) \quad (3)$$

where $h(A)$ is an adjustment factor

$$h(A) = (\tan^{-1}\{tr[I - B(A)]\})^{1/D} \quad (4)$$

a and $L(A)$ is a REML function

$$L(A) = |X_d' V^{-1} X_d|^{-1/2} |V|^{-1/2} \exp\left\{-\frac{1}{2} y' P \times \hat{\tau}_d^{DIR}\right\} \quad (5)$$

where $B(A) = \text{diag}(B_1, \dots, B_D)$, $B_d = \hat{v}(\hat{\tau}_d^{DIR}) / (A + \hat{v}(\hat{\tau}_d^{DIR}))$, $V = \text{diag}(A + \hat{v}(\hat{\tau}_d^{DIR}))$, $P = V^{-1} - V^{-1} X_d (X_d' V^{-1} X_d)^{-1} X_d' V^{-1}$, D is the number of domains (counties) in each state, X_d is a vector of auxiliary predictors, and $\hat{\tau}_d^{DIR}$ is a vector of the response variable, i.e. the domain-direct estimates for each county as in [1] and [2]. The adjusted maximum likelihood estimator of A is obtained by maximizing the adjusted likelihood $L_{adj}(A)$ with respect to A over $[0, \infty)$.

Model Selection and Fitting

In the R ‘sae’ package, the user specifies sample-direct estimates as the response variable in a mixed-effects linear regression model formula. We ran the procedure once with forest volume as the response, and again with forest AGB as the response. In addition to the county-level area coverage for each canopy height class, seven corresponding mountain region interaction terms (M*B5, M*B10, ..., M*B35) were included, with indicator M assigned unit values (M = 1) for counties in mountain areas of the states based on the FIA survey unit geography (Table 2). For counties in non-mountain regions, the indicator was assigned a value M = 0. Survey units are comprised of groups of counties within each state (Burrill et al. 2024). For example,

Table 2 Mountain codes based on the survey unit in FIA database. Mountain code value M = 1 is assigned to survey units in mountain regions, while M = 0 is for non-mountain survey units

State	Survey unit	Mountain code	Total number of counties	Mean volume (million m ³)	Mean biomass (million Mg)
NC	01 Southern Coast Plain	0	21	12.3	9.50
	02 Northern Coastal Plain	0	23	9.4	7.15
	03 Piedmont	0	35	11.1	8.17
	04 Mountains	1	21	18.4	14.10
TN	01 West	0	18	8.56	6.67
	02 West Central	0	11	11.5	9.97
	03 Central	0	23	7.35	6.07
	04 Plateau	0	16	12.4	10.30
	05 East	1	27	10.3	8.27
VA	01 Coastal Plain	0	34	8.34	6.36
	02 Southern Piedmont	0	17	14.7	11.50
	03 North Piedmont	0	17	11.08	9.15
	04 Northern Mountains	1	14	14.1	11.70
	05 Southern Mountains	1	17	13.8	10.90

in NC, a total of 21 counties are located in the Mountain survey unit (survey unit code 04, Table 2). In TN no survey unit is named for mountain geography; nonetheless, we designated the East survey unit as having mountain geography ($M = 1$) since its counties all lay within a subrange of the Appalachian Mountains. In VA two survey units named for northern and southern mountain units were coded with $M = 1$ (Table 2).

Besides specifying direct estimates as the response and CHM height class coverages as predictors in the regression model, county-level direct estimator variances $\hat{v}(\hat{\tau}_d^{DIR})$ were specified as inputs to the FH fitting procedure, as these are used for shrinkage calculation in the EBLUP estimation procedure. Separate fitting procedures were performed for each state to avoid the need for accounting for spatial covariances when fitting FH models to counties across multiple states (Cao et al. 2022). A zero-intercept regression model was chosen based on the intrinsic sense of having zero volume or AGB when all canopy height bin values were zero and as supported by preliminary comparisons of intercept versus no-intercept models using Akaike Information Criteria (AIC). The calculation of AIC in the ‘sae’ package was verified as appropriate for model selection based on REML estimates based on its use of the full-log-likelihood as noted by Verbyla (2019).

Variable selection was performed using all possible p -predictor subsets to identify best predictors while concurrently guarding against potential overfitting and large type-I error rates due to multiple comparisons. To these ends best model predictors for each combination of state, response variable, and remote-sensing data filter were determined based on models that had both the minimum AIC and all predictors’ statistical significance based on individual type-I error rates smaller than $\alpha = 0.01$.

Model Comparison

Thirty model scenarios were developed based on two response variables (forest volume and AGB), three states (NC, TN, and VA), and five combinations of CHM sources and filters (Fig. 1). In addition to meeting the minimum AIC criterion and having all predictors deemed significant based on $\alpha = 0.01$ we compared best model EBLUPs to direct sample estimates using a goodness of fit test following Brown et al. (2001). The test formulates differences between sample-direct estimates and FH EBLUPs, assuming a null hypothesis of no difference, specified as a Wald test

$$W(\hat{\tau}_d^{EBLUP}) = \sum_{i=1}^D \frac{(\hat{\tau}_d^{DIR} - \hat{\tau}_d^{EBLUP})^2}{\hat{v}(\hat{\tau}_d^{DIR}) + \widehat{MSE}(\hat{\tau}_d^{EBLUP})} \quad (6)$$

where $\hat{\tau}_d^{EBLUP}$ is the FH EBLUP for county domain d and $\widehat{MSE}(\hat{\tau}_d^{EBLUP})$ is the MSE for the FH EBLUP. The Wald test statistic [6] follows a χ^2 distribution with D degrees of freedom. Small values of the test statistic indicate good agreement between EBLUPS and direct estimates, while large values indicate disagreement. Precision of FH EBLUPs was assessed by the mean square error (MSE) calculated in the ‘sae’ package software as an additive combination of terms representing uncertainties associated with a) predicted random effects, b) estimation of regression

model coefficients, and c) estimation of the random effect variance, i.e. the variance among small-area domains (Molina and Marhuenda 2015b; Prasad and Rao 1990). The MSE is calculated as a second-order approximation using Taylor linearization, approximately unbiased when D is large. Approximate unbiasedness of the Prasad and Rao (1990) MSE estimator holds when any of three estimation techniques are used, including the package default REML used here.

County estimates and their standard errors from FIA sample-direct data were used as references to evaluate potential gains in precision from SAE. Relative standard errors for the direct estimates were calculated as

$$RSE\%_{DIR} = 100 * \sqrt{\hat{v}(\hat{\tau}_d^{DIR})} / \hat{\tau}_d^{DIR} \quad (7)$$

For comparison, relative mean-squared errors of FH EBLUP estimates were taken from ‘sae’ package outputs of approximated EBLUP MSEs

$$RMSE\%_{EBLUP} = 100 * \sqrt{MSE(\hat{\tau}_d^{EBLUP})} / \hat{\tau}_d^{DIR} \quad (8)$$

Preliminary analyses indicated that even for the smallest estimated $\hat{\tau}_d^{DIR}$ (0.32 for volume and 0.26 for AGB) $RSE\%$ and $RMSE\%$ values were not affected – none exceeded 102% – by numerical instability that can occur using ratio formulations with denominator values close to zero. As is customary when comparing design-unbiased sample-direct estimates, e.g., $\hat{\tau}_d^{DIR}$, to model-based SAE estimates subject to model biases, e.g., $\hat{\tau}_d^{EBLUP}$, we adopted terminologies “standard error” and “mean-squared error” in referring to design- versus model-based relative errors. A unitless standard error ratio (SER) for each county was calculated as

$$SER = RMSE\%_{EBLUP} / RSE\%_{DIR} \quad (9)$$

Random Effect Variance

Since the mixed models have two sources of variability (within and between-subjects), different types of residuals may be defined for corresponding analyses. Santos Nobre and da Motta Singer (2007) defined three types of residuals and indicated how each of them could be employed for diagnostics. Marginal residuals were defined as the difference between direct and synthetic estimates ($\hat{\tau}_d^{DIR} - X_d^T \hat{\beta}$), used to check linearity and within-county errors. Conditional residuals were calculated as differences between $\hat{\tau}_d^{DIR}$ and EBLUPs, used for evaluating homogeneity of variance across counties. Third are the random effects, which predict errors between the synthetic estimates and the EBLUPs, and are useful for checking normality of random effects variance, verifying equal distributions of random effects among different survey groupings, (e.g. survey units or ecological sections) and identifying outlying domains in the model population (Pinheiro and Bates 2000). To evaluate FH estimates of random effect variance among the counties, conditional residuals were used in log-likelihood calculations, comparing likelihoods for non-zero random effect variance (L_1) with likelihoods assuming zero random effect variance (L_2)

$$L_1 = (-0.5) * \sum_{i=1}^d \left(\ln(2 * \pi * (\hat{v}(\hat{\tau}_i^{DIR}) + A)) + \frac{(\hat{\tau}_i^{DIR} - \hat{\tau}_i^{EBLUP})^2}{(\hat{v}(\hat{\tau}_i^{DIR}) + A)} \right) \quad (10)$$

$$L_2 = (-0.5) * \sum_{i=1}^d \left(\ln(2 * \pi * \hat{v}(\hat{\tau}_i^{DIR})) + \frac{(\hat{\tau}_i^{DIR} - X_i^T \hat{\beta})^2}{\hat{v}(\hat{\tau}_i^{DIR})} \right) \quad (11)$$

Likelihood ratio tests (LRT) were then used to determine whether random effects improved models with synthetic estimates only,

$$LRT = -2 * (L_2 - L_1)$$

where L_1 and L_2 are the restricted log-likelihoods calculated as [10] and [11]. Under the null hypothesis a FH model with zero random effect variance, i.e. synthetic model, fits equally well as a FH model with positive random effect variance. Large values of the test statistic $LRT \sim \chi^2(1)$ provide evidence for concluding that the FH model with non-zero A improves on the fit over the regression-synthetic model lacking random effects (Marhuenda et al. 2016).

Results

Model Selection and Fitting

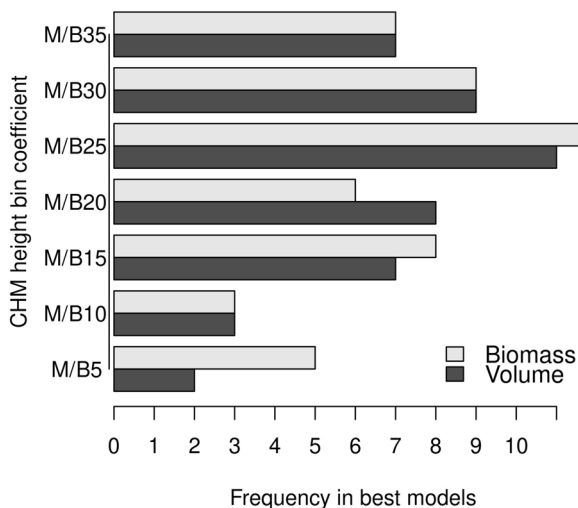
Model selection and fitting produced a set of 30 compact regression models for examining realized gains in forest inventory estimation by area-level SAE. Best FH models for county-specific EBLUP-based estimation of either AGB or wood volume in any state included between 2 and 4 predictors (mean = 3.2; mode = 3) with none of the best models supporting a regression intercept (Table 3). Conditional and marginal residual plots were examined to verify the suitability of model assumptions including the appropriateness of the linear model and stable variance of residuals across the range of observed values, none of which indicated any obvious violations (supplementary Figures S1– S90).

Predictors based on land areas (km^2) in the two shortest canopy height bins B5 and B10 (including both mountain and non-mountain counties) were selected in just 7 and 6 best models, respectively, of the 30 scenarios tested (Table 3, Fig. 2). All but two of these 13 occurrences occurred when NAIP-derived CHMs were used as auxiliary data. Statewide aerial extents for CHM bins B5-B35 showed greatest differences among filters tested in the B5 CHM bin, with a general trend of decreasing differences in larger CHM bins (Figs. 3 and 4). Of the three filters tested, the NAIP_noWater filter exhibited the greatest divergence above either GFCHM or NLCD filters, especially in the 5 m height bin (Fig. 4). The two scenarios where B5 predictors were selected for GFCHM auxiliary data both corresponded to models that estimated AGB, and both cases involved the NLCD filter

Table 3 Selected variables in best FH linear models for two forest attributes using five CHM source/filter combinations. Coefficient names starting with B denote CHM bins for all counties and MB coefficients apply additively to bins for mountain counties

State	Source/filter	Selected CHM bins	
		Volume	Biomass
NC	NAIP_noWater	(B10), B15, B30, B35	(B10), B15, B30, B35
	NAIP_GFCHM	(B10), B15, B30, B35	(B5), B15, B30, MB35
	NAIP_NLCD	B15, B30	B10, B20, B30
	GFCHM_default	B20, B25, MB30	B20, B25, B30
	GFCHM_NLCD	B20, B25, MB30	B5, B25, B30
TN	NAIP_noWater	B20, B35	B15, B25, MB35
	NAIP_GFCHM	B20, B35	B15, B25, MB35
	NAIP_NLCD	B20, B35	B15, B25, B35
	GFCHM_default	B25, B30, MB25	B20, B25, B30
	GFCHM_NLCD	B25, B30, MB25	B25, B30, MB35
VA	NAIP_noWater	B5, B15, B25, B35	B5, B15, B25, B35
	NAIP_GFCHM	B15, B25, MB10, (MB15)	B15, B25, MB10, (MB15)
	NAIP_NLCD	B5, B20, B35	B20, B35, MB5, (MB20)
	GFCHM_default	B20, B25, (MB25), MB30	B20, B25, MB30
	GFCHM_NLCD	B15, B25, (MB20), MB30	B5, B25, MB30

* parentheses indicate negative coefficient values

Fig. 2 Frequencies of selected predictors for CHM height bins B5-B35 including mountain interaction terms in best 15 FH regression models for biomass and volume attributes

(Table 3). Height bins for taller canopy classes (B15-B35) were all selected in ≥ 14 best model scenarios for combined volume and AGB response variables. In addition to the small number of models having significant B5 or B10 coefficients, 4 of their 13 occurrences resulted in negative coefficients, all for non-mountain

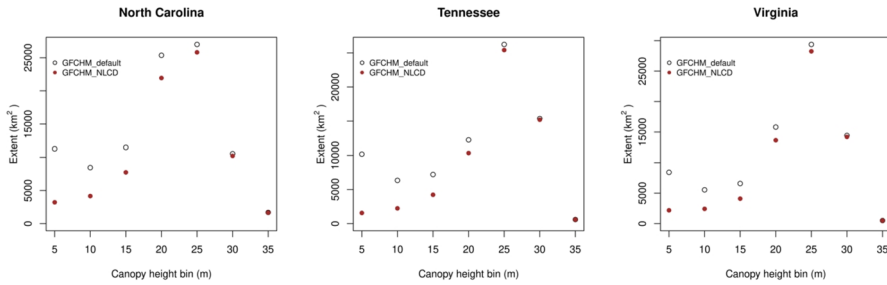


Fig. 3 Spatial extents of GFCHM canopy height bins for three states based on the default and NLCD filters

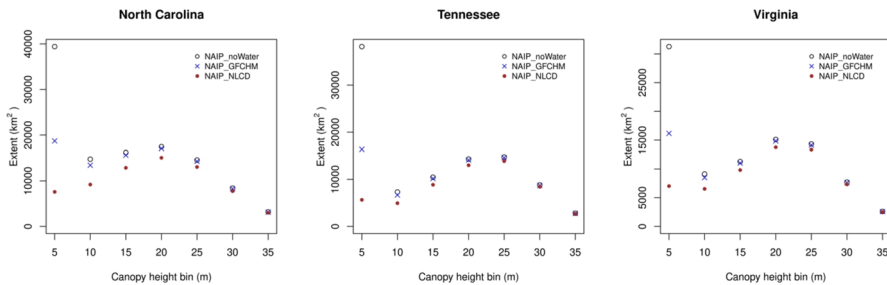


Fig. 4 Spatial extents of NAIP canopy height bins for three states based on no water, GFCHM, and NLCD filters

predictors and all involving NAIP-derived CHM predictors (see values in parentheses; Table 3).

Model coefficient units represent million cubic meters per square km ($10^6 \text{ m}^3/\text{km}^2$) for volume regression models and million metric tons per square km ($10^6 \text{ Mg}/\text{km}^2$) for AGB models (Table 4). Where best models for both response variables included common CHM bins, such as B20 and B25 in the NC GFCHM_default volume and AGB models (Tables 3 and 4, see also supplementary Table S1), coefficient ratios (AGB/Volume) should reflect the relative AGB in Mg (10^3 kg) per m^3 of wood growing in forest lands of three states. Ratios of FIA statewide estimated AGB to net wood volume from Table 1 were calculated as $990.7 \text{ Mg} \div 1,251 \text{ m}^3 = 0.79 \text{ Mg}/\text{m}^3$ for NC, $0.82 \text{ Mg}/\text{m}^3$ for TN, and $0.80 \text{ Mg}/\text{m}^3$ for VA, values we considered as gross averages for comparison to CHM bin coefficient ratios. Within-bin AGB/Volume coefficient ratios averaged $0.807 \text{ Mg}/\text{m}^3$ in 26 CHM bins common to the best volume and AGB models across the 15 state-by-CHM filter scenarios, with the individual scenario coefficient ratios exhibiting greater variation (range $[0.56, 1.23] \text{ Mg}/\text{m}^3$) compared to the narrow range of the three state-level ratios. We noted that values of AGB per m^3 of wood volume typically exceeded wood specific gravities of species common in these forests (Miles and Smith 2009). This result seemed sensible given that the definition of AGB includes wood, bark, and branch material in all trees having $\text{dbh} \geq 2.54 \text{ cm}$ (1 inch), while volumes only accounted for wood in stems having $\text{dbh} \geq 12.7 \text{ cm}$ (5 inch) and excluding stumps and stem topwood.

Table 4 Coefficient estimates for best FH linear models for two forest attributes using five CHM source/filter combinations. Coefficient names starting with B denote CHM bins for all counties and MB coefficients apply additively to bins for mountain counties

State	Data source	Response	Coefficients
NC	NAIP_noWater	Volume	B10 = -0.0232, B15 = 0.0659, B30 = 0.0345, B35 = 0.0612
NC	NAIP_GFCHM	Volume	B10 = -0.0249, B15 = 0.0684, B30 = 0.0316, B35 = 0.0656
NC	NAIP_NLCD	Volume	B15 = 0.0484, B30 = 0.0748
NC	GFCHM_default	Volume	B20 = 0.0181, B25 = 0.0189, MB30 = 0.027
NC	GFCHM_NLCD	Volume	B20 = 0.0198, B25 = 0.0179, B30 = 0.0298
TN	NAIP_noWater	Volume	B20 = 0.0421, B35 = 0.1024
TN	NAIP_GFCHM	Volume	B20 = 0.0428, B35 = 0.1035
TN	NAIP_NLCD	Volume	B20 = 0.0459, B35 = 0.107
TN	GFCHM_default	Volume	B25 = 0.0231, B30 = 0.015, MB25 = 0.0079
TN	GFCHM_NLCD	Volume	B25 = 0.0236, B30 = 0.0153, MB25 = 0.0089
VA	NAIP_noWater	Volume	B5 = 0.0047, B15 = 0.0327, B25 = 0.03, B35 = 0.0699
VA	NAIP_GFCHM	Volume	B15 = 0.0301, B25 = 0.0538, MB10 = 0.1256, MB15 = -0.0881
VA	NAIP_NLCD	Volume	B5 = 0.0352, B20 = 0.0479, B35 = 0.0826
VA	GFCHM_default	Volume	B20 = 0.0139, B25 = 0.0283, MB25 = -0.0149, MB30 = 0.0225
VA	GFCHM_NLCD	Volume	B15 = 0.0811, B25 = 0.0247, MB20 = -0.0429, MB30 = 0.0185
NC	NAIP_noWater	Biomass	B10 = -0.0183, B15 = 0.0511, B30 = 0.0292, B35 = 0.0416
NC	NAIP_GFCHM	Biomass	B5 = -0.0074, B15 = 0.0442, B30 = 0.039, MB35 = 0.0262
NC	NAIP_NLCD	Biomass	B10 = 0.0222, B20 = 0.0227, B30 = 0.0498
NC	GFCHM_default	Biomass	B20 = 0.0145, B25 = 0.0126, B30 = 0.0214
NC	GFCHM_NLCD	Biomass	B5 = 0.0918, B25 = 0.0159, B30 = 0.022
TN	NAIP_noWater	Biomass	B15 = 0.0137, B25 = 0.0368, MB35 = 0.0401
TN	NAIP_GFCHM	Biomass	B15 = 0.0142, B25 = 0.0371, MB35 = 0.0401
TN	NAIP_NLCD	Biomass	B15 = 0.0233, B25 = 0.0262, B35 = 0.06
TN	GFCHM_default	Biomass	B20 = 0.0117, B25 = 0.0129, B30 = 0.0164
TN	GFCHM_NLCD	Biomass	B25 = 0.0197, B30 = 0.0129, MB25 = 0.0058
VA	NAIP_noWater	Biomass	B5 = 0.0034, B15 = 0.0275, B25 = 0.0228, B35 = 0.0558
VA	NAIP_GFCHM	Biomass	B15 = 0.0249, B25 = 0.0404, MB10 = 0.0888, MB15 = -0.0579
VA	NAIP_NLCD	Biomass	B20 = 0.0523, B35 = 0.0675, MB5 = 0.0305, MB20 = -0.0179
VA	GFCHM_default	Biomass	B20 = 0.015, B25 = 0.0184, MB30 = 0.0102
VA	GFCHM_NLCD	Biomass	B5 = 0.0938, B25 = 0.0202, MB30 = 0.0103

Comparing CHM Sources and Forest Cover Filters

Comparison of AIC statistics for a given state and response variable show some differences in model efficacy (i.e., smallest AIC) based on the choice of CHM source (GFCHM versus NAIP) and forest/non-forest filters (GFCHM or NLCD; Table 5). While variable selection was determined in part by minimum AIC values for individual models, AIC was also used to make comparisons between best models for pairs of CHM source and forest/non-forest filter combinations, namely: 1) comparing NAIP_GFCHM vs. GFCHM_default sources, which both employ filtering that

Table 5 Akaike Information Criteria (AIC) and Wald test chi-squared statistics* from Fay-Herriot best models

Measure	Source	Volume			Biomass		
		NC	TN	VA	NC	TN	VA
AIC	NAIP_noWater	441.4	417.1	459.8	379.2	369.6	403.4
	NAIP_GFCHM	441.6	416.8	454.7	379.0	369.5	396.8
	NAIP_NLCD	439.1	418.0	446.2	379.4	368.5	398.9
	GFCHM_default	445.0	423.7	448.7	383.1	369.9	405.1
	GFCHM_NLCD	436.6	424.5	447.4	377.2	372.3	404.2
Wald test χ^2	NAIP_noWater	47.34	24.34	25.18	50.90	25.22	28.19
	NAIP_GFCHM	70.10	24.66	27.54	66.70	25.18	32.51
	NAIP_NLCD	72.49	25.96	32.00	69.94	25.77	33.74
	GFCHM_default	55.39	23.08	29.49	64.34	23.82	27.53
	GFCHM_NLCD	67.58	23.37	33.10	59.41	24.21	27.44

*Wald test χ^2 critical values ($\alpha=.05$) for NC, TN, and VA are 124.3, 118.8, and 123.2 respectively, based on numbers of counties in each state

integrates LANDSAT and GEDI information (Potapov et al. 2021); and 2) comparing NAIP_NLCD vs. GFCHM_NLCD sources, which are both filtered based on NLCD forest cover classification. Our AIC results showed the potential for additional – perhaps modest – improvement over the NAIP_noWater approach in several scenarios we examined by alternative filtering based on either GFCHM or NLCD information (NC volume and VA volume and AGB; Table 5).

Comparisons between GFCHM_default and NAIP_GFCHM CHM sources showed absolute values of AIC differences >2 for all three states' volume models and for two of the three states' AGB models – excepting Tennessee – with the threshold of 2 adopted as an approximate critical value for significant evidence ($\alpha=0.05$) to reject a “no-difference” null hypothesis between competing models. When comparison involved NAIP and GFCHM CHMs filtered by NLCD forest classification, i.e. NAIP_NLCD vs. GFCHM_NLCD, AIC was only unable to distinguish between FH volume models for Virginia (Table 5).

Because competing models can't be clearly distinguished when their AICs differ by <2 , comparison of CHM sources and data filters was aided by examining standard error reductions of FH EBLUP RMSEs over direct estimate standard errors, i.e. SER (Eq. [9]; Fig. 5). In NC and VA volume and biomass estimates augmented by NAIP CHM auxiliary information gave smaller SERs in FH EBLUPs when either of the forest/non-forest filters (NAIP_GFCHM or NAIP_NLCD) was used (Fig. 5). Distinguishing between the two filters was not always clear in VA and NC, but the use of one or the other over the simpler “no water” filter was generally supported for NAIP-derived CHMs by both AIC and SER comparisons. For TN models involving NAIP CHMs, neither SERs nor AIC provided clear guidance for whether the NAIP_noWater filter was outperformed by either of the other two CHM filters (Fig. 5). The NLCD filter resulted in significant improvement using GFCHM auxiliary data only in NC (GFCHM_default vs. GFCHM_NLCD; Table 5).

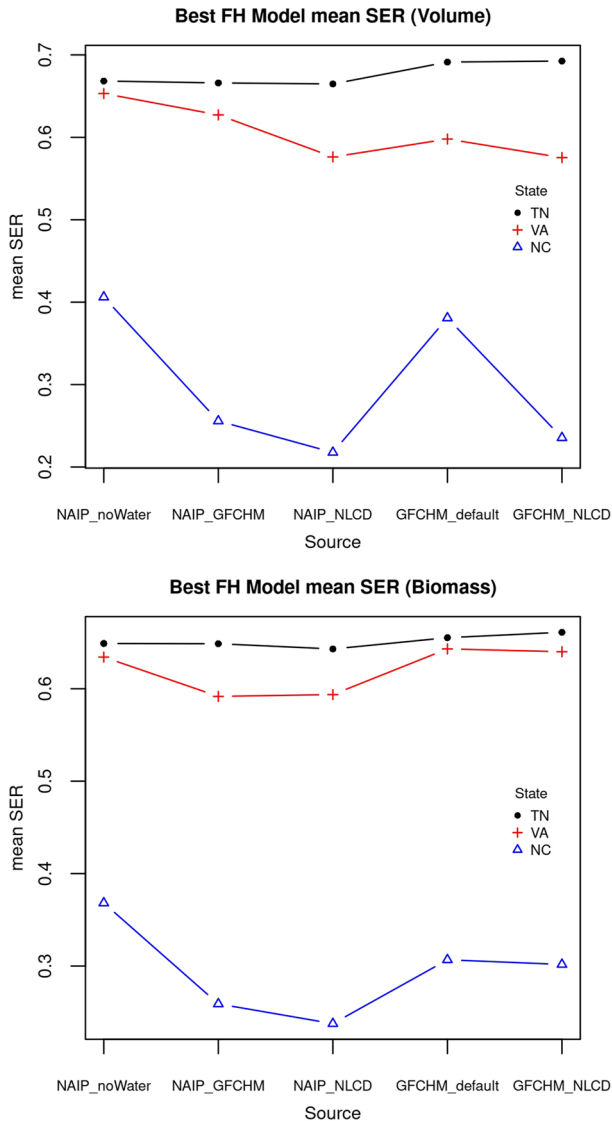


Fig. 5 Fay-Herriot (FH) EBLUP RMSE to direct sample standard error ratios (SER; Eq. [9]) averaged across counties from estimated forest volume (**a**) and aboveground biomass (**b**)

In contrast, unfiltered GFCHM data gave detectable but only slightly better results for modeling biomass in TN compared to GFCHM_NLCD filtered data (Fig. 5, Table 5). Filtering CHMs from either source by masking non-forested areas had variable results, with no apparent gains in TN models over those based on NAIP_noWater and GFCHM_default auxiliary data sets. Although AIC showed no significant improvement in NC biomass models when either NAIP_GFCHM or NAIP_NLCD filters were used, SER results showed substantial reductions in EBLUP for the same models.

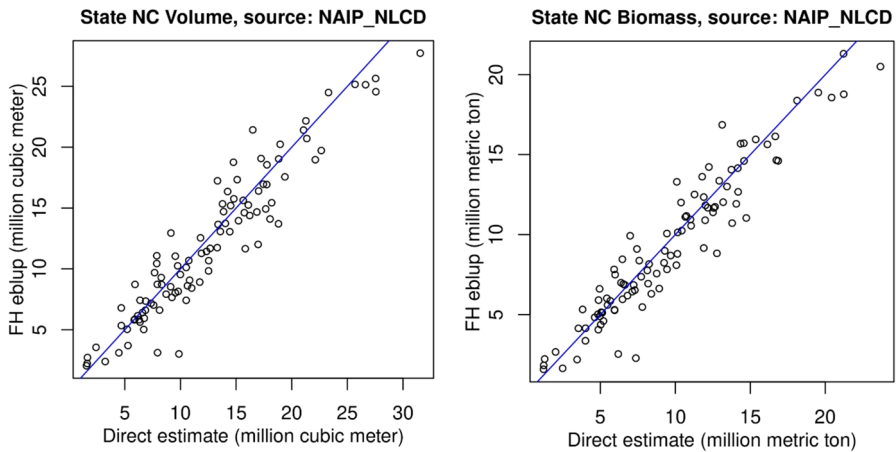


Fig. 6 Fay-Herriot (FH) composite estimates (EBLUP) of county volume (left) and biomass (right) using NAIP_NLCD sources in North Carolina compared to corresponding FIA direct estimates

Looking at NLCD mask effects on GFCHM data, AIC indicated that the NLCD mask improved model performance compared to using GFCHM_default data for both NC volume and AGB models. Notwithstanding, these gains were only reflected by SER in NC for volume, not AGB. Results for AIC and SER seemed more consistent with each other comparing CHM filters in VA, where drops in $AIC > 2$ from one CHM filter scenario to another coincided with noticeable drops in SER. Despite the lack of consistent patterns for making generalizations about which CHM data source and forest/non-forest filter combinations performed best as auxiliary data in FH models studied here, the range of SERs from [0.22, 0.69] across all combinations of CHM and filters confirmed the ability of FH estimators to deliver significant gains in precision for estimating county-level forest volume and AGB (supplementary Table S1).

Fay-Herriot County-Level Estimates

Wald test χ^2 statistics did not indicate any significant lack-of fit for the FH county-level EBLUP estimates of Volume or AGB in any of the scenarios examined (Table 5). These tests also served as a check for possible EBLUP inconsistencies with direct estimates that might indicate bias in FH models. Graphical comparison of county-level direct estimates to FH EBLUPs also showed no obvious inconsistencies that might point to biases in FH estimates (Fig. 6). The NAIP_NLCD results presented in Fig. 6 were chosen based having the largest χ^2 test statistics of the scenarios tested, even though none of the Wald test results exceeded critical value thresholds (Table 5).

Random Effect Variances

After recalculating negative random effect variances from the ‘sae’ package’s default REML estimator with the adjusted maximum likelihood-based formula [3], estimates

Table 6 Random effect variance of forest volume and aboveground biomass attributes among counties, and log-likelihood-ratio-test ρ values (in parentheses) for a null of “zero random effect variance” in FH models ($\chi^2 \sim 1$ degrees of freedom)

Source	Volume ($\text{m}^3 \times 10^6$) ²			Biomass (metric ton $\times 10^6$) ²		
	NC	TN	VA	NC	TN	VA
NAIP_noWater	0.5998 (< 0.001)	2.7040 (< 0.001)	3.1619 (< 0.001)	0.2259 (0.0011)	1.4922 (< 0.001)	1.6236 (< 0.001)
NAIP_GFCHM	0.0329 (0.2494)	2.6651 (< 0.001)	2.6792 (< 0.001)	0.0202 (0.2267)	1.4892 (< 0.001)	1.2362 (< 0.001)
NAIP_NLCD	0.0335 (0.2347)	2.6448 (< 0.001)	2.0332 (< 0.001)	0.0176 (0.2513)	1.4346 (< 0.001)	1.2579 (< 0.001)
GFCHM_default	0.5213 (< 0.001)	3.0304 (< 0.001)	2.2221 (< 0.001)	0.1197 (0.0073)	1.5507 (< 0.001)	1.7690 (< 0.001)
GFCHM_NLCD	0.0249 (0.3092)	3.0553 (< 0.001)	1.9279 (< 0.001)	0.1120 (0.0089)	1.6047 (< 0.001)	1.7342 (< 0.001)

of A ranged from $[0.03, 3.16] (\text{m}^3 \times 10^6)^2$ for volume and $[0.02, 1.77] (\text{Mg} \times 10^6)^2$ for AGB (Table 6). Likelihood ratio tests indicated five models with random effect variances near the low ends of these ranges showed little to no evidence ($p > 0.05$) that random effects in the FH estimator improved on a fixed-effects-only regression-synthetic estimator (Table 6). County-level EBLUPs reflected the minimal impact of random effects in adjusting composite estimators away from synthetic estimates and toward the survey direct estimates in models when LRTs showed no evidence of support for random effects, e.g., NC AGB with NAIP_NLCD (Fig. 7, left panel). In contrast, where LRTs showed strong evidence for including random effect, the effect of EBLUPs in adjusting synthetic estimates toward the direct estimator was considerable, e.g., VA AGB with GFCHM_default (Fig. 7, right panel). We note here for clarity that when FH EBLUPS lie closer to the line $Y = X$ than synthetic-model predictions in plots like those shown in Fig. 7, the differences are due to random effects, which act to move the predictions closer toward the line representing equivalence with direct estimates.

Discussion

The use of binned CHM height classes as predictors was effective in formulating linear models to characterize relationships between auxiliary data and forest volume and AGB attributes for model fitting. While biomass models involving lidar metrics as predictors have been shown to require transformations to achieve linearity and stability of residual variance (Duncanson et al. 2022; Hansen et al. 2017; May et al. 2023), the suitability of linear models here was advantageous in eliminating the need for non-linear regression analysis or transformations requiring back-transformation correction factors to reduce transformation biases (Clifford et al. 2013; Paul et al. 2018).

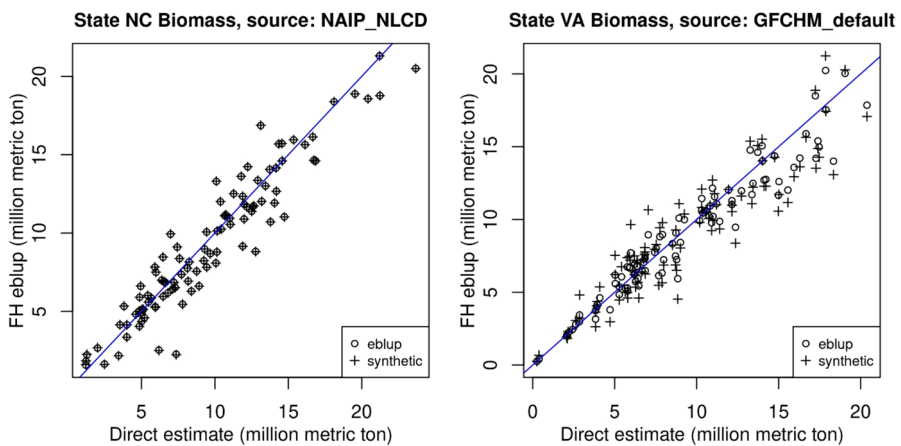


Fig. 7 Comparison between EBLUPS, synthetic-model predictions and direct estimates of county-level biomass in North Carolina (left) using NAIP_NLCD CHM data, and Virginia biomass (right) using GFCHM_default CHM auxiliary data

Selection of CHM bins as predictors indicated that area of land covered by relatively low forest canopies (≤ 10 m) was generally not as useful in predicting forest volume or AGB in the forests studied; nevertheless, B5 and B10 coefficients did significantly contribute to some best models in states of NC and VA. When NAIP CHMs served as auxiliary data, B5 and B10 bins appeared in best models for both volume and AGB attributes. When GFCHMs were used as auxiliary data, the B5 predictor was chosen only in two best models for AGB.

In addition to results indicating fewer meaningful relationships between area covered by low-canopy forests and wood volume or AGB attributes, there appeared to be some justification for further examination of NAIP canopy heights ≤ 10 m to determine the cause of apparent negative relationships between canopy heights and forest wood volume or biomass. Prior et al. (2022), for example, noted that shadows in steep terrain caused NAIP-derived CHMs to overestimate dominant tree heights. Errors of this kind in DAP-derived point clouds or CHMs could account for negative regression coefficients such as the ones we noted, although the occurrence of negative coefficients we observed in B5 and B10 height bins did not arise in mountain-region counties. In contrast, negative valued coefficients for a few scenarios in CHM bins MB15-MB25 did point to a possible over-representation of volume or biomass by binned CHM predictors for some height classes in mountain regions. Similar to shadow-caused errors in steep terrain noted by Prior et al. (2022), Potapov et al. (2021) reported some overestimation of vegetation heights in steep slopes, which they made efforts to adjust for. Further study could help determine whether such overestimates might relate to negative regression coefficients for mountain-region counties observed in some FH models developed here. Although the negative coefficients raised the possibility of regression fixed-effects models predicting negative volume or AGB, no negative estimates were observed for counties in this study based on either EBLUPs or fixed-effects model predictions.

Whereas Cao et al. (2022) demonstrated the utility of using NAIP-derived CHMs filtered to remove water bodies, our AIC results show the potential for additional – perhaps modest – improvement over the NAIP_noWater approach in some scenarios we examined by additional filtering based on either GFCHM or NLCD information (Table 5). We note here that the GFCHM of Potapov et al. (2021) accounted for areas classified as non-tree canopy by setting their gridded values to zero, which were not tabulated in the lowest (B5 or MB5) CHM bin predictors used here. Transferring GFCHM zero values to NAIP CHMs in our NAIP_GFCHM filter showed mixed results. While AIC indicated gains were only achieved in VA by applying the NAIP_GFCHM filter in place of NAIP_noWater, we noted at least modest gains (reductions) in SERs for both NC and VA using NAIP_GFCHM. We also noted some modest improvements in modeling FIA volume and AGB using the NLCD forest classification filter. We attribute this to slight differences in the definitions of forest, forest canopy, or forest land adopted by the US FIA program and other national or global institutions (cf. Tables 1, 2, 16 in Nelson et al. 2020, USDA Natural Resources Conservation Service 2015).

A practical concern is the strong performance of the GFCHM in the FH modeling context despite its 30 m grid resolution compared to the finer 10 m resolution of NAIP CHMs tested here. Previous work has shown that resampling raster auxiliary

data from 1 to 10 m or 30 m does not necessarily degrade the precision of FH estimates in forest inventory (Green et al. 2020b). In other work Green et al. (2022) obtained FH EBLUPs to estimate stand-level green weights in pine plantations and mixed pine-hardwood forests in Alabama, Mississippi, and Washington using 10 m resolution Sentinel-2 imagery both with and without aerial lidar height metrics as auxiliary data. Although we did not attempt to resample NAIP CHMs to a coarser 30 m resolution for comparison to GEDI-derived GFCHMs, it seems plausible that for counties like those in NC, TN, and VA, which average 1220 km² in land area, 30 m resolution CHMs could be relied on to give adequate results.

The use of passive-optical sensor satellite data is known to be subject to problems such as underestimation of AGB-related parameters due to saturation of spectral reflectance, especially in forest sites with high biomass density (Lu et al. 2016; Mutanga et al. 2023). Although this concern was anticipated for the GFCHM data tested here due to its reliance on Landsat multispectral data, its use as auxiliary data in estimating county-level forest volume and biomass in the three eastern states we studied compared favorably with NAIP-derived DAP CHMs. Problems with DAP methods have been reported including underestimation of tree and canopy heights, especially where shadows in canopy gaps or thinning corridors, steeply sloping terrain, or other anomalies interfere with photogrammetric solutions (Ritz et al. 2022; Schroeder et al. 2022). Based on the graphical comparison of county-level direct estimates to FH EBLUPS (Figs. 6 and 7) and the Wald χ^2 test statistics (Table 5), there did not appear to be any apparent biases related to oversaturation of spectral reflectance.

Notwithstanding concerns with either of the CHM data sources we tested, similar precision gains in the ranges of those achieved here (Fig. 5) have previously been reported in studies that linked NFI data to DAP-derived metrics (Breidenbach et al. 2018; Magnussen et al. 2017). Ver Planck et al. (2018) showed greater gains with SERs as low as 0.19 in an application involving landscape-scale forest inventory data paired with aerial lidar. Green et al. (2022) reported SERs between 0.69 and 0.9 for FH estimates of live tree green weights in forest stands ranging from about 7 ha to 117 ha. While the choice of auxiliary data sources, direct sampling parameters such as plot size and sampling intensity, and spatial scales of domains of interest can greatly affect gains of FH and related SAE methods over direct estimates alone, the satellite and DAP-derived CHMs used here compared favorably to a range of applications that have combined field survey data and auxiliary data, with the latter most-often derived from aerial or satellite-based remote sensing (Coulston et al. 2021; Goerndt et al. 2011, 2019; Green et al. 2020a; Mauro et al. 2017; Temesgen et al. 2021). Although not strongly indicated for state-specific, county-level models like those developed here, past work has shown additional gains in precision by accounting for spatial autocorrelation among model residuals (Magnussen et al. 2017; Ver Planck et al. 2018), and by allowing greater flexibility in the specification of heteroskedasticity in model variance across the range of predictions (Breidenbach et al. 2018).

Because FH composite estimators adjust design-unbiased survey direct estimates toward regression synthetic model estimates, they do not preserve the design-unbiased property of survey-direct estimates. Nonetheless, the composite nature of FH

EBLUPs serves to guard against the introduction of large model biases. Statistical properties of FH EBLUPs and the widely demonstrated robustness of FH estimators lend a degree of confidence in estimates' unbiasedness even though it is not assured by survey design and direct estimator properties (Jiang and Rao 2020; Lahiri and Rao 1995; Molina et al. 2015). As with any model-based estimators, checking model assumptions is essential to support their basis for asymptotic properties, such as asymptotic unbiasedness of FH estimates and approximate unbiasedness of EBLUP variances (Dettmann et al. 2022; Prasad and Rao 1990; Rao and Molina 2015).

An implication of zero (or negative) estimated random effect variance in FH models is the potentially unwanted or unmerited over-shrinkage to the regression-synthetic estimator, even for area domains having moderately large sample sizes (Yoshimori and Lahiri 2014). Tests for zero random effect variance may alleviate some of this concern and may serve to justify the choice of regression synthetic estimators rather than composite FH models, thus obtaining more precise estimates without undue concern over model biases (Marhuenda et al. 2016, Santos Nobre and da Motta Singer 2007). While concerns over bias in regression-synthetic estimators must be considered in deciding whether the FH approach is suitable for a particular application, ongoing research and the availability of demonstrated solutions should serve to assist analysts in addressing problems of this nature (Datta et al. 2011).

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s44391-025-00021-z>.

Data Availability Data supporting this study is included within the article as supplementary information.

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Authors and Affiliations

**Qianqian Cao¹ · Philip J. Radtke¹ · John W. Coulston² · Valerie A. Thomas¹ ·
Randolph H. Wynne¹ · David M. Walker³**

✉ Qianqian Cao
qianqian21@vt.edu

Philip J. Radtke
pradtke@vt.edu

John W. Coulston
john.coulston@usda.gov

Valerie A. Thomas
thomasv@vt.edu

Randolph H. Wynne
wynne@vt.edu

David M. Walker
david.walker@usda.gov

¹ Department of Forest Resources and Environmental Conservation, Virginia Tech, Blacksburg, USA

² Southern Research Station, USDA Forest Service, Blacksburg, VA, USA

³ Southern Research Station through Oak Ridge Institute for Science and Education, USDA Forest Service, Blacksburg, VA, USA